Technology

Miniature magnet rivals behemoths in strength

Karmela Padavic-Callaghan

FOR the first time, a magnet small enough to fit in your palm can match the strength of some of the world's most powerful magnets.

Strong magnets are used in everything from magnetic resonance imaging to nuclear fusion. The most powerful are made from superconductors, materials that conduct electricity with near-perfect efficiency. But superconducting magnets that produce strong magnetic fields are often bulky, with the largest comparable to a two-storey building, says Alexander Barnes at ETH Zurich in Switzerland.

He and his colleagues have built a superconducting magnet that is competitive with large magnets in strength, but measures only 3.1 millimetres in diameter (*Science Advances*, doi.org/qvgp). They made it by coiling a thin tape of a ceramic material called REBCO, which superconducts when cooled to extremely low temperatures. These coils produce magnetic fields when electric currents pass through them.

The team bought the REBCO tape from a commercial company, then set out to find the best magnet design, making and testing over 150 of them, says Barnes.

Their chosen design involves either two or four pancake-shaped coils of REBCO that could produce magnetic fields with strengths of 38 tesla and 42 tesla, respectively. For comparison, a fridge magnet usually has a strength under 0.01 T. The two magnets that produce the world's strongest steady magnetic fields reach around 45 T, weigh many tonnes and require up to 30 megawatts of power. This team's magnet can fit in your hand and uses less than 1 watt of power.

Mark Ainslie at King's College London says this "suggests that extremely high-field magnets could become more accessible to a wider range of laboratories in the near future". ▮

Health

Smell test could help us spot Parkinson's disease sooner

Chris Simms

PEOPLE with Parkinson's disease seem less able to enjoy pleasant smells, such as lemon. This has led scientists to conclude that "the world smells different" with the condition. The discovery could help doctors cheaply and non-invasively diagnose Parkinson's, a process that usually takes several years.

Loss of the sense of smell is a core symptom of Parkinson's, affecting 75 to 90 per cent of people and often preceding the tremors commonly associated with the condition. There have been many efforts to use the loss of smell as a diagnostic tool, but these have been complicated by the fact that this sense also declines with healthy ageing.

Now, Noam Sobel at the Weizmann Institute of Science in Rehovot, Israel, and his team have tried a different approach: testing smell perception.

They recruited 94 people, most of whom were in their late 50s to late 60s. Thirty-three of them had been diagnosed with Parkinson's, while another 33 had no known medical conditions, and 28 had smell dysfunction unrelated to Parkinson's. The researchers used standard tests and questionnaires to assess participants' ability to detect and identify smells.

"Anything that brings us closer to helping identify someone's personal risk would be great"

They also used their own tests to find what they call an olfactory perceptual fingerprint. These involved asking the participants to rate the intensity and pleasantness of smells emanating from three jars. One contained a high concentration of the odorant citral, which smells like lemon, the next had a mix of the aromatic ingredients asafoetida and skatole that was so concentrated it smelled faecal, and the third was an empty jar.

All the tests detected when the participants had a decline in their ability to sense smells, but only the olfactory perceptual fingerprints could distinguish between people with smell loss who did or didn't have Parkinson's, telling the two groups apart with 88 per cent accuracy. This rose to 94 per cent when the age and sex of the participants were matched.

Those with Parkinson's perceived the citrus smell as being just as intense as the healthy group did, and more intense than those with smell problems unrelated to the condition. Yet when it came to pleasantness, both groups with smell issues scored lower than the healthy one. The people with Parkinson's also sniffed nearly 2 per cent longer in response to the unpleasant smell compared with the lemon one, whereas those in the other groups cut their sniff duration by 11 to 12 per cent (medRxiv, doi.org/qvgn).

Sobel and his team speculate that smelling functions correctly in the nose of someone with Parkinson's, but that their brain processes the signals differently, resulting in a sniffing response that no longer relates to how pleasant a scent is.

This is probably associated with changes in brain areas such as the anterior olfactory nucleus, which shrinks when deprived of smell signals, and is suspected to be one of the earliest sites of Parkinson's brain pathology.

Michał Pieniak at the Smell & Taste Clinic at the Dresden University of Technology in Germany says of every 10 people who come into the clinic having lost their sense of smell and a reason can't be found, approximately one will develop Parkinson's. "Anything that would bring us closer to helping identify their personal risk would be great." ▮

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Lemons may smell more pleasant to people without Parkinson's disease